



Twistors

&

Anomalies

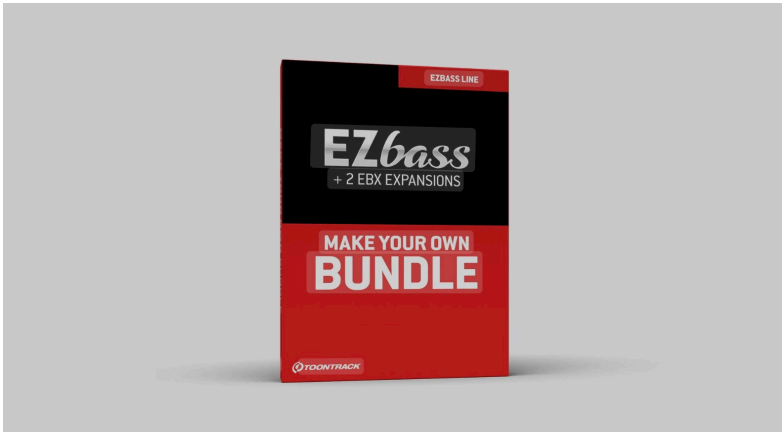


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Line Bundles



① Line Bundles

A line bundle over $\mathbb{P}^1$ of degree n is depicted as $\mathcal{O}(n)$. The fibres of this bundle are rational functions of coordinates on $\mathbb{P}^1$.

The idea is that when we go from the holomorphic patch to anti-holomorphic patch; i.e. $\bar{z} = \frac{1}{z}$ then for $(u(z), z) \in \mathcal{O}(n)$

the fibre coordinate $u(z) = \frac{u(\bar{z})}{\bar{z}^n} = z^n u(\bar{z})$

Example 1: Constant functions are fibres of $\mathcal{O}(0)$ bundle

Example 2: Consider $(u, v, 1, z) \in \mathbb{CP}^3$ such that

$(1, z) \in \mathbb{CP}^1$ and u, v do not depend on z

$$(u, v, 1, z) = z \left(\frac{u}{z}, \frac{v}{z}, \frac{1}{z}, 1 \right) \approx \left(\frac{u}{z}, \frac{v}{z}, \bar{z}, 1 \right)$$

In the anti-holomorphic patch $\bar{u} = \frac{u}{z}$, $\bar{v} = \frac{v}{z}$

$\Rightarrow u, v$ fibrate $\mathcal{O}(1)$ and $(u, v, 1, z)$ belongs to the bundle

$$\mathcal{O}(1) \oplus \mathcal{O}(1) \rightarrow \mathbb{CP}^1$$

When we pull-back to the space of forms the definition changes. A (p,q) valued differential form $U(z)$ is a fibre of the $\mathcal{O}(n)$ bundle if:

$$U(z) = \frac{U(\bar{z})}{z^n} = \bar{z}^n U(\bar{z}) \in \Omega^{p,q}$$

Note: It is a common practice to call the fibre coordinate $U(z)$ itself as the $\mathcal{O}(n)$ bundle.

Example 3: The canonical bundle (Top holomorphic form) $K_{\mathbb{P}^1}$ of $\mathbb{CP}^1$ is $\mathcal{O}(-2)$.

$$z = \frac{1}{\bar{z}} \Rightarrow dz = \frac{-d\bar{z}}{\bar{z}^2} = \frac{-d\bar{z}}{z^{-2}}$$

$$\Rightarrow dz \in K_{\mathbb{P}^1} \cong \mathcal{O}(-2)$$

Twistor Geometry



② Twistors

Flat Twistor space $\mathbb{PT} := \mathbb{CP}^3 \setminus \mathbb{CP}^1$
is the space of spinors of the
complexified conformal group $SL(4, \mathbb{C})$.

In a nice representation, we can write

$$Z^\alpha = (\mu^{\dot{A}}, \lambda_A) \in \mathbb{PT}$$

such that $\mu^{\dot{A}}$ and λ_A are left and right -
handed spinors of the complexified
Minkowski space $M_{\mathbb{C}}$ respectively such that

$$\mu^{\dot{A}} = x^{A\dot{A}} \lambda_A ; x^{A\dot{A}} \in M_{\mathbb{C}} \text{ \& } \lambda_A \in \mathbb{CP}^1$$

Another interesting perspective is to
consider twistors as trivial fibrations of $\mu^{\dot{A}}$
over λ_A ($\mathbb{CP}^1$).

let's say $\mu^{\dot{A}} = (\mu^{\dot{0}}, \mu^{\dot{1}}) = (u, v)$

$$\lambda_A = (1, z) \in \mathbb{CP}^1$$

then $\mathcal{Z}^{\dot{A}} = (\mu^{\dot{A}}, \lambda_A) = (u, v, 1, z) \in \mathbb{CP}^3 \setminus \mathbb{CP}^1$

$$\Rightarrow \mathcal{Z}^{\dot{A}} = z \left(\frac{u}{z}, \frac{v}{z}, \frac{1}{z}, 1 \right) \approx \left(\overset{\bar{u}}{\underbrace{\frac{u}{z}}}, \overset{\bar{v}}{\underbrace{\frac{v}{z}}}, \bar{z}, 1 \right)$$

$\Rightarrow u, v$ are $\mathcal{O}(1)$ bundles over $\mathbb{CP}^1$.

$$\Rightarrow \mathbb{PT} \cong \mathcal{O}(1) \oplus \mathcal{O}(1) \rightarrow \mathbb{CP}^1.$$

A nice thing about this perspective is that spacetime points will then be the sections of this bundle.

$$\mathcal{M}_{\mathbb{C}} \hookrightarrow \mathcal{O}(1) \oplus \mathcal{O}(1) \rightarrow \mathbb{CP}^1.$$

Let's try to see this,

we know from a spacetime perspective that

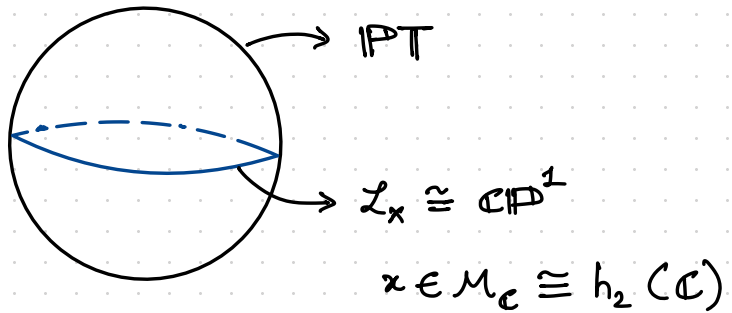
$$x^{AA} = \frac{1}{\sqrt{2}} \begin{bmatrix} t-z & x+iy \\ x-iy & t+z \end{bmatrix} \in \overset{\rightarrow 2 \times 2 \text{ hermitian}}{\mathfrak{h}_2(\mathbb{C})}$$

but a fixed spacetime point x^{AA} on $\mathbb{PT}$ satisfies $\mu^{\dot{A}} = x^{AA} \lambda_A \quad \forall (\mu^{\dot{A}}, \lambda_A) \in \mathbb{PT}$

∴ It is a section in the bundle $\mathcal{O}(1) \oplus \mathcal{O}(1) \rightarrow \mathbb{CP}^1$

Moreover, the incidence relation $\mu^{\dot{A}} = x^{A\dot{A}} \lambda_A$ is describing a 2-plane $\mathbb{C}^2 \subset \mathbb{C}^4 \cong \mathbb{T}$. The projective version of this 2-plane is $\mathbb{CP}^1$.

∴ Spacetime points are Riemann spheres (or lines) in Twistor space

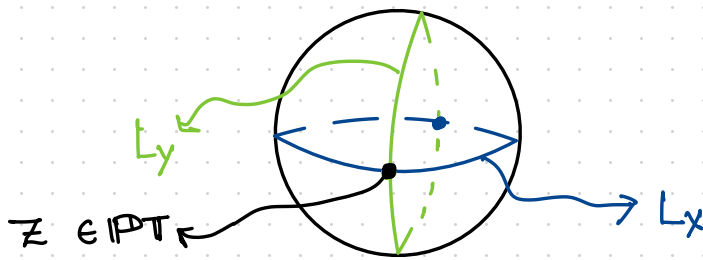


Consider two "twistor lines" L_x and L_y that intersect at some point $(\mu^{\dot{A}}, \lambda_A)$ in $\mathbb{PT}$.

We have,

$$\mu^{\dot{A}} = x^{A\dot{A}} \lambda_A$$

$$\mu^{\dot{A}} = y^{A\dot{A}} \lambda_A$$



Two lines L_x and L_y intersecting in $\mathbb{PT}$

It follows that $(x^{A\dot{A}} - y^{A\dot{A}}) \lambda_A = 0 \Rightarrow x - y \propto \lambda$

$$\Rightarrow x^{A\dot{A}} - y^{A\dot{A}} = \tilde{\lambda}^{\dot{A}} \lambda^A \text{ for some } \tilde{\lambda}^{\dot{A}}$$

$$\det(\tilde{\lambda}^{\dot{A}} \lambda^A) = 0 \Rightarrow x^{A\dot{A}} - y^{A\dot{A}} \text{ is null.}$$

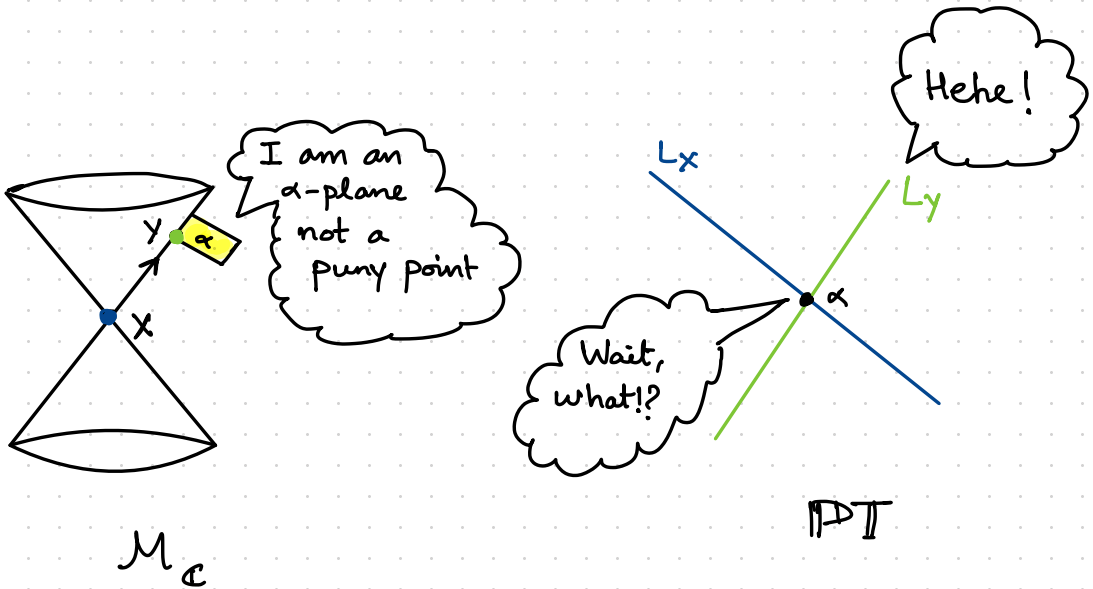
•• Lines in twistor intersect only if the corresponding spacetime points are null separated

The point of intersection $Z \in \mathbb{PT}$ is parametrized by $\tilde{\lambda}^{\dot{A}}$ in spacetime which is complex 2-plane often referred to as the α -plane.

Summary:

- Lines in twistor space $\longleftrightarrow$ points in spacetime
- Points in twistor space $\longleftrightarrow$ α -planes in spacetime

Every thing is twisted !



Bitwistors and spacetime points

We have seen that spacetime points correspond to lines in twistor space. But a line can be described by 2-points that lie on it!

Let X be a spacetime point and Z_1^α, Z_2^β be two points that lie on L_X . Then the bitwistor

$X^{\alpha\beta} = Z_1^{[\alpha} Z_2^{\beta]}$ is given by:

$$X^{\alpha\beta} = \langle \lambda_1, \lambda_2 \rangle \begin{bmatrix} \frac{1}{2} \epsilon^{\dot{A}\dot{B}} x^2 & x_B^{\dot{A}} \\ -x_A^{\dot{B}} & \epsilon_{AB} \end{bmatrix}$$

$\therefore$ The bitwistor $X^{\alpha\beta}$ encodes all the information about the spacetime point upto a proportionality factor of $\langle \lambda_1, \lambda_2 \rangle = \epsilon_{AB} \lambda_1^A \lambda_2^B$

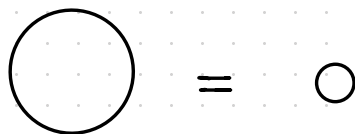
! Note:

Beautiful
correspondence with ←
Klein-quadric

This is what Woit calls Grassmanian 2-plane!

${}^4C_2 = 6 \rightarrow$ vector of Conformal space (complexified)

Conformal structure



A natural $sl(4, \mathbb{C})$ invariant scalar product on twistor space is

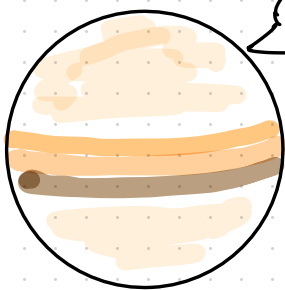
$$\langle z_1, z_2, z_3, z_4 \rangle = \epsilon_{\alpha\beta\gamma\delta} z_1^\alpha z_2^\beta z_3^\gamma z_4^\delta$$

One can check that for Bitwistors $X^{\alpha\beta}$ and $X^{\gamma\delta}$

$$\begin{aligned} \epsilon_{\alpha\beta\gamma\delta} dX^{\alpha\beta} dX^{\gamma\delta} &= \langle \lambda, \lambda \rangle^2 dx^{AA} dx_{AA} \\ &= \langle \lambda, \lambda \rangle^2 ds^2 \end{aligned}$$

But $Sl(4, \mathbb{C})$ transformations are conformal transformations and there is no difference between ds^2 and $\langle \lambda, \lambda \rangle^2 ds^2$!

To break conformal symmetry, we need to fix the asymptotics of our theory which is achieved by choosing an infinity twistor.



Do you know
about the
largest planet?

I don't care
as long as
they are
all round!

$Sl_4(\mathbb{C})$

Fun fact: An ∞ -twistor is not a twistor!
(It's a bi-twistor)

We will modify our metric to:

$$ds^2 = \frac{\epsilon_{\alpha\beta\gamma\delta} dX^{\alpha\beta} dX^{\gamma\delta}}{(I_{\alpha\beta} X^{\alpha\beta})^2}$$

where $I_{\alpha\beta}$ is the infinity twistor.

Ex: Consider $I_{\alpha\beta} = \frac{1}{2} \begin{bmatrix} 0 & 0 \\ 0 & \epsilon_{AB} \end{bmatrix}$

then $I_{\alpha\beta} X^{\alpha\beta} = \langle \lambda, \lambda_2 \rangle \frac{\epsilon^{AB}}{2} \epsilon_{AB} = \langle \lambda, \lambda_2 \rangle$

$\Rightarrow ds^2 = (dx^{AA} dx_{AA}) \rightarrow$ Asymptotically flat !!

I guess; $\frac{1}{2} \begin{bmatrix} \Lambda \epsilon^{AB} & 0 \\ 0 & \epsilon_{AB} \end{bmatrix}$ gives ds or AdS !

Penrose Transforms



SD-ASD decomposition

Any 2-form field can be written as:

$$F_{A\dot{A}B\dot{B}} = \epsilon_{AB} \underbrace{F_{\dot{A}\dot{B}}}_{\text{self dual}} + \epsilon_{\dot{A}\dot{B}} \underbrace{F_{AB}}_{\text{anti-self dual}}$$

In 4d $\rightarrow$ $*$ of a 2-form is a 2-form

$(*)^2 = 1$ in Euclidean signature
 $\downarrow$

eigen-values $\begin{cases} +1 & \text{SD} \\ -1 & \text{ASD} \end{cases}$

$$F = \frac{1}{2} \underbrace{(F + *F)}_{\text{SD}} + \frac{1}{2} \underbrace{(F - *F)}_{\text{ASD}}$$

In vacuum, Maxwell's equation + Bianchi identity

become:

$$\partial^A{}_{\dot{B}} F_{AB} + \partial^{\dot{A}}{}_{\dot{B}} F_{\dot{A}\dot{B}} = 0$$

$$\partial^A{}_{\dot{B}} F_{AB} - \partial^{\dot{A}}{}_{\dot{B}} F_{\dot{A}\dot{B}} = 0$$

$$\Rightarrow \partial^{\dot{A}\dot{B}} F_{AB} = 0 \quad \& \quad \partial^{\dot{A}\dot{B}} F_{\dot{A}\dot{B}} = 0$$

Similarly, the Riemann tensor can be decomposed as:

$$R_{ABCD} \dot{A} \dot{B} \dot{C} \dot{D} = \epsilon_{AB} \epsilon_{CD} C_{\dot{A} \dot{B} \dot{C} \dot{D}} + \epsilon_{\dot{A} \dot{B}} \epsilon_{\dot{C} \dot{D}} C_{ABCD} + \epsilon_{\dot{A} \dot{B}} \epsilon_{CD} \Phi_{\dot{A} \dot{B} \dot{C} \dot{D}} \\ + \epsilon_{AB} \epsilon_{\dot{C} \dot{D}} \Phi_{\dot{A} \dot{B} CD} + \frac{R}{12} (\epsilon_{AB} \epsilon_{CD} \epsilon_{\dot{A} \dot{D}} \epsilon_{\dot{B} \dot{C}} \\ + \epsilon_{AD} \epsilon_{BC} \epsilon_{\dot{A} \dot{B}} \epsilon_{\dot{C} \dot{D}})$$

where $C_{\dot{A} \dot{B} \dot{C} \dot{D}} \rightarrow$ SD Weyl tensor

$C_{ABCD} \rightarrow$ ASD Weyl tensor

$\Phi \rightarrow$ trace-free Ricci tensor; $R \rightarrow$ Ricci scalar.

Einstein eqⁿ in vacuum $\Rightarrow R=0$; $\Phi=0$.

Bianchi identity

$$\epsilon_{CD} \partial_B^{\dot{A}} C_{\dot{A} \dot{B} \dot{C} \dot{D}} - \epsilon_{\dot{C} \dot{D}} \partial_B^{\dot{A}} C_{ABCD} = 0$$

For SD Weyl Gravity $C_{ABCD} = 0$

$$\text{so } \partial_B^{\dot{A}} C_{\dot{A} \dot{B} \dot{C} \dot{D}} = 0$$

likewise for ASD Weyl gravity $C_{\dot{A} \dot{B} \dot{C} \dot{D}} = 0$

$$\Rightarrow \partial_B^{\dot{A}} C_{ABCD} = 0$$

In general for field with spin s and helicity h

$$h=1 \quad \partial_{\dot{B}}^{\dot{A}_1} \phi_{\dot{A}_1 \dot{A}_2 \dots \dot{A}_{2s}} = 0$$

$$h=0 \quad \square \phi = 0$$

$$h=-1 \quad \partial_{\dot{B}}^{\dot{A}_1} \phi_{\dot{A}_1 \dot{A}_2 \dots \dot{A}_{2s}} = 0$$

where $\phi_{\dot{A}_1 \dots \dot{A}_{2s}}$ captures the SD trace-free anti-symmetric part of the curvature tensor.

Theorem (Penrose): Zero-rest mass chiral fields on spacetime correspond to homogenous holomorphic (if on-shell) functions on twistor space.

Negative helicity $h = -1$, spin s field

$$\Phi_{A_1 \dots A_{2s}}(x) = \int_{L_x \cong \mathbb{CP}^1} \langle \lambda d\lambda \rangle \lambda_{A_1} \lambda_{A_2} \dots \lambda_{A_{2s}} f(z)$$

The field on the L.H.S. is not homogenous in $\lambda_A \in \mathbb{CP}^1$.

So, the integrand must have homogeneity 0 in λ .

Everything other than $f(z)$ is homogeneous

$2s+2$, so $f(z)$ should have homogeneity $-2s-2$.

$\langle \lambda d\lambda \rangle$ is $(1,0)$ form, $\therefore$ to integrate over $L_x \cong \mathbb{CP}^1$

f should be a $(0,1)$ form.

$$\Rightarrow f \in \Omega^{0,1}(\mathbb{P}^1, \mathcal{O}(-2s-2))$$

If the equation of motion is true then

$$\partial_B^{A_1} \phi_{A_1 \dots A_{2s}} = 0$$

$$\Rightarrow \int_{L_x} \langle \lambda d\lambda \rangle \lambda_{A_1} \dots \lambda_{A_{2s}} \left[\lambda^{A_1} \frac{\partial f}{\partial \mu^{\dot{B}}} + \bar{\lambda}^{A_1} \frac{\partial f}{\partial \bar{\mu}^{\dot{B}}} \right] \Big|_{L_x} = 0$$

The first term $\lambda^{A_1} \frac{\partial f}{\partial \mu^{\dot{B}}}$ is 0 since

$$\lambda_{A_1} \lambda^{A_1} = 0.$$

$\therefore$ For the whole thing to be zero $\frac{\partial f}{\partial \bar{\mu}^{\dot{B}}} = 0$

$\Rightarrow f(z)$ is holomorphic.

$$\bar{\partial} f = 0$$

It is a bit more effort to show that if

$$f = \bar{\partial} g \text{ then } \phi_{A_1 \dots A_{2s}} = 0$$

$$\therefore \bar{\partial} f = 0 ; f \neq \bar{\partial} g \Rightarrow f \in H^{0,1}(\mathbb{PT}, \mathcal{O}(-2s-2))$$

f is in the first Dolbeault cohomology class if ϕ is on-shell.

Positive helicity $h=1$, spin s field

$$\Phi_{\dot{A}_1 \dots \dot{A}_{2s}}(x) = \int_{L_x} \langle \lambda d\lambda \rangle \frac{\partial}{\partial \mu^{\dot{A}_1}} \frac{\partial}{\partial \mu^{\dot{A}_2}} \dots \frac{\partial}{\partial \mu^{\dot{A}_{2s}}} f(z)$$

The integrand other than $f(z)$ is homogenous $-2s+2$ in λ . Hence $f(z)$ should be homogenous $2s-2$.

$$\Rightarrow \boxed{f(z) \in \Omega^{0,1}(\mathbb{PT}, \mathcal{O}(2s-2))}$$

By the same arguments as before, if $\Phi_{\dot{A}_1 \dots \dot{A}_{2s}}$ is on-shell, then

$$\boxed{f \in H^{0,1}(\mathbb{PT}, \mathcal{O}(2s-2))}$$

Gauge theories

(Ward - Correspondence)

SD Yang-Mills
Instanton Moduli
space in $\mathbb{R}^4$.

$\longleftrightarrow$

Holomorphic $\mathcal{O}(0)$
gauge valued bundles
on $\mathbb{P}\mathbb{T}$.

$$\begin{array}{c} s=1 \\ \uparrow \\ 2(1)-2 \\ \uparrow \end{array}$$

The complex structure on $\mathbb{P}\mathbb{T}$ is essential to describe the $\bar{\partial}$ operator.

We can deform the complex structure to get an almost complex structure J st $J^2 = -1$.

In this case, the differential operator becomes

$$\bar{\partial} \rightarrow \bar{\partial} + a = \bar{D} \quad \text{for some } a \in \Omega^{0,1}(\mathbb{P}\mathbb{T})$$

The almost complex structure becomes a complex structure if

$$\bar{D}^2 = \bar{\partial}a + \frac{1}{2}[a, a] = F \in \Omega^{0,2}(\mathbb{P}\mathbb{T}) = 0$$



A little more effort is needed to show that $F \in \Omega^{0,2}(PT, \mathcal{O}(1)) = 0$ actually corresponds to $F_{AB} = 0$ which is SD Yang-Mills.

$$F_{AB} \tilde{A} \tilde{B} = \epsilon_{AB} F_{\tilde{A} \tilde{B}}$$

Lights, Camera, &
Action



BF Action (Gauge Theory)

Consider the action

$$S = \int_{\mathbb{P}T} b \wedge F \quad ; \quad a \in \Omega^{0,1}(\mathbb{P}T, \overset{\substack{\text{adjoint} \\ \uparrow}}{g})$$

$$\text{where } F = \bar{\partial} a + \frac{1}{2} [a, a] \in \Omega^{0,2}(\mathbb{P}T, g)$$

$$= \bar{D}^2, \text{ where } \bar{D} = \bar{\partial} + a$$

$$b \in \Omega^{3,1}(\mathbb{P}T, g^*)$$

g^* is the dual of g w.r.t. the Killing-Cartan metric.

Note that $\Omega^{3,0}$ is the canonical bundle $K_{\mathbb{P}T}$ over $\mathbb{P}T$ for some chart $(u, v, 1, z) \in \mathbb{P}T$

$$\Omega^{3,0} = du dv dz$$

$$dz \rightarrow \text{canonical bundle of } \mathbb{CP}^1 \cong \mathcal{O}(-2)$$

$$du \rightarrow \frac{d\bar{u}}{\bar{z}^{-1}} \cong \mathcal{O}(-1)$$

$$dv \rightarrow \frac{d\bar{v}}{\bar{z}^{-1}} \cong \mathcal{O}(-1)$$

$$\Rightarrow K_{\mathbb{P}T} \cong \mathcal{O}(-4)$$

∴ the b field can be written as

$$b \in \Omega^{0,1}(\mathbb{P}^1, \mathcal{O}(-4))$$

note: $\mathcal{O}(-4)$ is also $\mathcal{O}(-2(1) - 1)$,
so b corresponds to a gauge field with
negative helicity.

Integrating out the Lagrange multiplier gives

$$F = 0$$

which is the integrability condition for the
complex structure $\bar{D}$.

It also implies SD Yang-Mills solutions F_{AB} .

Integrating out the a field gives

$$\bar{D}b = 0$$

Which is a holomorphic helicity -1 field
propagating freely in the SD Yang-Mills background.

Just like Yang-Mills; B-F theory is invariant under the following gauge transformation:

$$a \rightarrow g(\bar{D})g^{-1} ; \quad b \rightarrow g b g^{-1}$$

$$\Rightarrow bF \rightarrow g b g^{-1} g F g^{-1} = bF$$

infinitesimally

$$\delta a = \bar{D}c + [a, c] = \bar{D}c ; \quad \delta b = [b, c] .$$

But BF theory is invariant also under

$$\delta b = \bar{D}d ; \quad \delta a = 0$$

This follows from Bianchi identity

$$\bar{D}F = 0 .$$

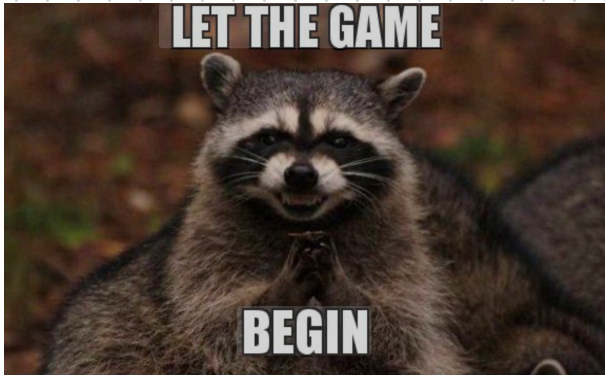
In total, it is invariant under

$$\delta a = \bar{D}c ; \quad \delta b = \bar{D}d + [b, c]$$

$$\forall b, c \in \Omega^{0,0}(\mathbb{P}^1, g)$$

SD Conformal Gravity v/s

SD Einstein Gravity



Conformal gravity action in spacetime is given by

$$S = \int_{M^4} d^4x \sqrt{g} \left[\Psi_{ABCD} \Psi^{ABCD} + \Psi_{\dot{A}\dot{B}\dot{C}\dot{D}} \Psi^{\dot{A}\dot{B}\dot{C}\dot{D}} \right]$$

where Ψ_{ABCD} is ASD Weyl tensor

We can subtract to this the topological term

$$\int d^4x \sqrt{g} \left[\Psi_{ABCD} \Psi^{ABCD} - \Psi_{\dot{A}\dot{B}\dot{C}\dot{D}} \Psi^{\dot{A}\dot{B}\dot{C}\dot{D}} \right]$$

to get

$$S = \int_{M^4} d^4x \sqrt{g} \left[\Psi_{\dot{A}\dot{B}\dot{C}\dot{D}} \Psi^{\dot{A}\dot{B}\dot{C}\dot{D}} \right] \rightarrow \text{SD conformal gravity?}$$

This is ~~not~~ the SD conformal gravity action!

The solution of the Conformal gravity action is the vanishing of the Bach tensor;

$$\begin{aligned} B_{\mu\nu} &= 2 \nabla^\rho \nabla^\sigma C_{\rho\sigma\mu\nu} + C_{\rho\sigma\mu\nu} R^{\rho\sigma} \\ &= 2 (\nabla_{\dot{A}}^{\dot{C}} \nabla_{\dot{B}}^{\dot{D}} + \Phi_{\dot{A}}^{\dot{C}} \Phi_{\dot{B}}^{\dot{D}}) \Psi_{ABCD} ; \\ &= 2 (\nabla_{\dot{A}}^{\dot{C}} \nabla_{\dot{B}}^{\dot{D}} + \Phi_{\dot{A}}^{\dot{C}} \Phi_{\dot{B}}^{\dot{D}}) \Psi_{\dot{A}\dot{B}\dot{C}\dot{D}} ; \quad \Phi \rightarrow \text{traceless Ricci tensor} \end{aligned}$$

The ~~Correct~~ action for SD - conformal gravity is:

$$S = \int_M d^4x \sqrt{g} \quad G_{ABCD} \Psi^{ABCD}$$

And the Equations of motion are:

$$\Psi_{ABCD} = 0 \quad \text{and} \quad (\nabla_{\dot{A}}^{\dot{C}} \nabla_{\dot{B}}^{\dot{D}} + \Phi_{\dot{A}}^{\dot{C}} \Phi_{\dot{B}}^{\dot{D}}) G_{ABCD} = 0$$

This is a BF theory action with G_{ABCD} being an ASD spin 2 field propagating in the background of SD Weyl tensor.

SD Einstein gravity action is :

$$S = \int_M g_{ABCD} R^{ABCD}$$

where g_{ABCD} is the ASD part of the tetrad
two-form $e \wedge e$

and R^{ABCD} is the ASD part of the curvature of
the spin Connection

$$R = d\omega + \frac{1}{2}[\omega, \omega] = \bar{D}^2; \quad \bar{D} = d + \omega$$

Equations of motion are:

$$R^{ABCD} = 0 \quad \& \quad \bar{D} g_{ABCD} = 0.$$

Note: Both SD conformal gravity and SD Einstein gravity have a BF theory type action!

Next, let us review how these actions look on
Twistor space.

The Non-linear Graviton construction (Penrose):

There is a correspondance between curved manifolds with a conformal class of metrics $[g]$ and complex 3-folds $\mathbb{P}\mathcal{Y}$ that are complex deformations of $\mathbb{P}\mathcal{T}$ with at least one rational curve $\mathbb{L}_x$ with normal bundles $\mathcal{O}(1) \oplus \mathcal{O}(1)$.

The complex deformation is captured by

$$\bar{\nabla} = \bar{\partial} + \mathcal{L}_f \rightarrow \text{Lie derivative along } f$$

flat background

$$f \in \Omega^{0,1}(\mathbb{P}\mathcal{Y}, T_{\mathbb{P}\mathcal{T}})$$

f is a $(0,1)$ form valued in the tangent space of flat twistor space.

$$f = f^\beta_{\alpha} d\bar{z}^\alpha \frac{\partial}{\partial z^\beta}$$

Note that is background dependent. However in conformal gravity one can rescale the metric. Einstein gravity is genuinely background dependent.

conformal gravity is more natural on twistor space and we will discuss this before Einstein gravity

The almost complex structure is integrable if the Nijenhuis tensor vanishes

$$N^\alpha = \bar{\partial} f^\alpha + \frac{1}{2} [f \wedge f]^\alpha \in \Omega^{0,2}(\mathbb{P}\mathcal{Y}, T_{\mathbb{P}\mathcal{T}})$$

$$\text{for } f^\alpha = f^\alpha_{\bar{\beta}} d\bar{z}^{\bar{\beta}} \in \Omega^{0,1}(\mathbb{P}\mathcal{Y}, T_{\mathbb{P}\mathcal{T}})$$

$$[f \wedge f]^\alpha = f^\beta \wedge \partial_{\bar{\beta}} f^\alpha \quad (\text{Lie derivative for vectors})$$

The vanishing of the Nijenhuis tensor imposes SD Weyl tensor as we shall see from this action:

$$S = \int_{\mathbb{P}\mathcal{Y}} G_{\alpha} \wedge \left(\bar{\partial} f^\alpha + \frac{1}{2} [f \wedge f]^\alpha \right) \quad [\text{Mason 2005}]$$

$$\begin{aligned} \text{where } G_{\alpha} &\in \Omega^{3,1}(\mathbb{P}\mathcal{Y}, T_{\mathbb{P}\mathcal{T}}^*) \\ &\cong \Omega^{0,1}(\mathbb{P}\mathcal{Y}, K_{\mathbb{P}\mathcal{T}} \otimes T_{\mathbb{P}\mathcal{T}}^*) \end{aligned}$$

The equations of motion imply

$$N^\alpha = 0 \quad \text{and} \quad \bar{\partial} G_{\alpha} + [f \wedge G]_{\alpha} = \bar{\nabla} G_{\alpha} = 0$$

This is SD conformal gravity on twistor space with an ASD negative helicity Graviton G_{α} propagating freely.

To go from SD conformal gravity to SD Einstein gravity we will have to fix the conformal structure; i.e.; asymptotics of our theory. This is done by choosing an infinity twistor.

Recall that for flat space the ∞ twistor is given by

$$I_{\alpha\beta} = \begin{pmatrix} 0 & 0 \\ 0 & \epsilon_{\dot{A}\dot{B}} \end{pmatrix}$$

Therefore we need to preserve the symplectic two-form $d\mu_{\dot{A}} \wedge d\mu^{\dot{A}}$ on the fibres of our curved twistor space to preserve the infinity twistor.

This makes the game much more easier since now we know that $\mathbb{P}\mathbb{Y}$ is a $\mathcal{O}(1) \oplus \mathcal{O}(1)$ fibre bundle over $\mathbb{CP}^1$ just like in flat twistor space.

The complex deformations are given by:

$$\bar{\nabla} = \bar{\partial} + \mathcal{L}_v$$

where $v \in \Omega^{0,1}(\mathbb{P}\mathbb{Y}, T_{\mathbb{P}\mathbb{T}})$

$$v = v^{\alpha}_{\bar{\beta}} d\bar{z}^{\beta} \frac{\partial}{\partial \bar{z}^{\alpha}}.$$

But more importantly this complex deformation

preserves the symplectic 2-form (or equivalently the Poisson bracket) on the fibres and therefore

$$V = \epsilon^{\dot{A}\dot{B}} \frac{\partial h}{\partial \mu^{\dot{A}}} \frac{\partial}{\partial \mu^{\dot{B}}} = \{h, \cdot\}$$

where $h \in \Omega^{0,1}(\mathbb{P}\mathcal{Y}, T_{\mathbb{P}\mathcal{T}})$

Another convenience is that $T_{\mathbb{P}\mathcal{T}}$ can be replaced by $\mathcal{O}(2)$ due to existence of the fibration $\mathbb{P}\mathcal{Y} \rightarrow \mathbb{CP}^1$.

Similarly $T_{\mathbb{P}\mathcal{T}}^*$ can be replaced by $\mathcal{O}(-2)$.

The Mason-Wolf action for SD Conformal gravity now becomes the Sharma action for SD Einstein gravity:

$$S = \int_{\mathbb{P}\mathcal{Y}} g \wedge (\bar{\partial}h + \frac{1}{2} \{h, h\})$$

where $\bar{\partial}h + \frac{1}{2} \{h, h\} = T \in \Omega^{0,2}(\mathbb{P}\mathcal{Y}, \mathcal{O}(2))$

$$g \in \Omega^{3,1}(\mathbb{P}\mathcal{Y}, \mathcal{O}(-2)) \cong \Omega^{0,1}(\mathbb{P}\mathcal{Y}, \mathcal{O}(-6))$$

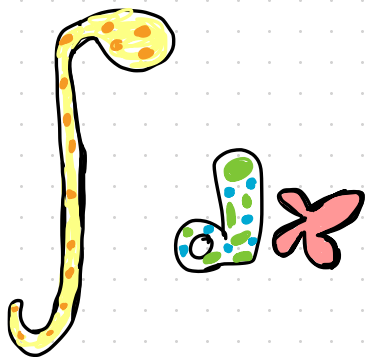
Notice that $2 = 2(2) - 2$ is the positive helicity graviton whereas $-6 = -2(2) - 2$ is the negative helicity graviton.

The equations of motion for this action are:

$$T=0 \quad \text{and} \quad \bar{\partial}g + \{h, g\} = \bar{\nabla}g = 0$$

The vanishing of T also makes the Nijenhuis tensor vanish making $\bar{\nabla}$ integrable

$$N = (\bar{\partial} + \{h, \cdot\})^2 = \{T, \cdot\} \hat{=} 0$$



All this drama
just to be integrable!

Gauge transformations (Diffeomorphisms):

So far we have seen that SD conformal gravity, SD Einstein gravity, and SD Yang-Mills all have very similar looking actions.

It is easy to guess what diffeomorphisms look like based on gauge transformations of SD Yang Mills.

$$\delta h = \bar{\partial} \chi + \{h, \chi\} = \bar{\nabla} \chi,$$

$$\delta g = \{g, \chi\} + \bar{\partial} \phi + \{h, \phi\} = \{g, \chi\} + \bar{\nabla} \phi$$

$$\forall \chi \in \Omega^{0,0}(\mathbb{P}^3, \mathcal{O}(2)); \phi \in \Omega^{0,0}(\mathbb{P}^3, \mathcal{O}(-6)).$$

Note: h has weight $\mathcal{O}(2)$ but $\{.,.\} = \varepsilon^{\dot{A}\dot{B}} \frac{\partial}{\partial \mu^{\dot{A}}} \frac{\partial}{\partial \mu^{\dot{B}}}$ has weight $\mathcal{O}(-2)$ and $\therefore \{h, .\}$

does not have a weight.

SD conformal gravity is also invariant under

$$\delta f = 0$$

$$\delta G = \bar{\partial} \phi + [f, \Phi]; \quad \phi \in \Omega^{0,0}(\mathbb{P}^3, T_{\mathbb{P}^3}^* \otimes K_{\mathbb{P}^3})$$

$\delta f = \bar{\partial} \chi + [f, \chi]$ makes sense for $\chi \in \Omega^{0,0}(\mathbb{P}^3, T_{\mathbb{P}^3})$ but the compensating term $\delta G = [G, \chi]$ doesn't?

$Sl(2, \mathbb{C})$ v/s $Gl(3, \mathbb{C})$ connections on $T_{\mathbb{P}^3}$.

The complex deformations $\bar{\nabla} = \bar{\partial} + L_v$ deform the tangent space as. There are $Sl(2, \mathbb{C})$ connections for SD Einstein tangent space but $Gl(3, \mathbb{C})$ connections for SD conformal gravity tangent space.

Let's see: $\bar{\nabla}(u)$ for $u \in T_{\mathbb{P}^3}$ is

$$\begin{aligned} & \bar{\partial}(u) + L_v(u) \\ &= \bar{\partial} u^\alpha \partial_\alpha + (L_v u^\alpha) \partial_\alpha + u^\alpha L_v(\partial_\alpha) \\ &= (\bar{\nabla} u^\alpha) \partial_\alpha - (u^\beta \partial_\beta v^\alpha) \partial_\alpha \end{aligned}$$

$\Rightarrow$ The connection induced by the complex deformation is $S_\beta^\alpha = -\partial_\beta v^\alpha \rightarrow Gl(3, \mathbb{C})$ connections in general.

for SD Einstein $v^\alpha = v^{\dot{A}} = \partial^{\dot{A}} h$

the connection for the normal bundle to $\mathbb{CP}^1$ appears when $b = \dot{B}$

$$S_{\dot{B}}^{\dot{A}} = -\partial_{\dot{B}} \partial^{\dot{A}} h \rightarrow \text{these are } Sl(2, \mathbb{C}) \text{ connections.}$$

But in general in the conformal case they will be $Gl(3, \mathbb{C})$ connections.

Gauge Fixing

Let's take the BF action for SD Yang-Mills.

$$S = \int b \wedge \bar{D}a$$

Define $\bar{D}^+ = * \bar{D} * : \Omega^{p,q}(PT) \rightarrow \Omega^{p,q-1}(PT)$

Faddeev-Popov gauge fixing for YM requires adding a BRST exact term to the action $S\Psi$

where $\Psi = \bar{c}(\bar{D}^+a - m)$

$$S\Psi = m\bar{D}^+a - m^2 + \bar{c}\bar{D}^+Dc$$

Integrating out the Nakanishi-Lautrup field m gives

$$S\Psi = \frac{1}{2}(\bar{D}^+a)^2 + \bar{c}\bar{D}^+Dc.$$

The beauty of BRST is that Ψ can be anything, as long as we add BRST exact $S\Psi$ to the action there is no issue. Instead of doing Faddeev-Popov one can take

$$\Psi = \bar{c}\bar{D}^+a ; S\Psi = m\bar{D}^+a + \bar{c}\bar{D}^+Da.$$

now integrating out m directly fixes the gauge $\bar{D}^+a = 0$.

$\therefore$ The gauge fixed action can be

$$S_{gf} = \int b \bar{D} a + m \bar{\partial}^+ a + n \bar{\partial}^+ b + \bar{c} \bar{\partial}^+ \bar{D} c + \bar{e} \bar{\partial}^+ \bar{D} e$$

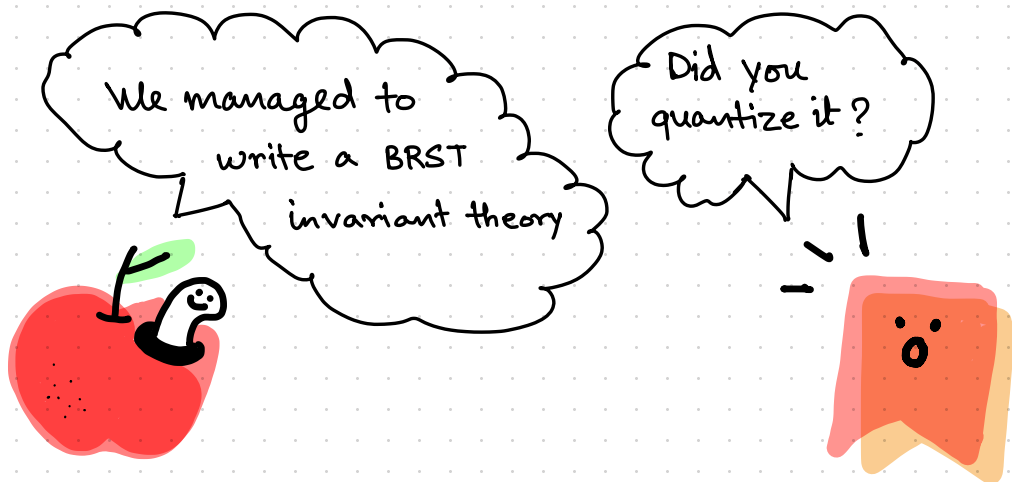
$c \in \Omega^0(g)$ and $e \in \Omega^{3,0}(g)$ are ghost fields of a and b

$\bar{c} \in \Omega^{3,3}(g)$ and $\bar{e} \in \Omega^{0,3}(g)$ are anti-ghost fields

$m \in \Omega^{3,3}(g)$ and $n \in \Omega^{0,3}(g)$ are Nakanishi
Lautrup fields.

Important: Gauge Symmetry is not a symmetry!

The actual global symmetry of the gauge fixed
action is BRST symmetry.



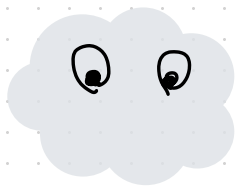
Even if the classical action is BRST invariant;
the effective action W given by:

$$e^W = Z = \int [D\alpha D\beta \dots] e^{-S}$$

may not be BRST invariant.

$$\delta_B S = 0 \text{ but } \delta_B W = \text{Anomaly!}$$

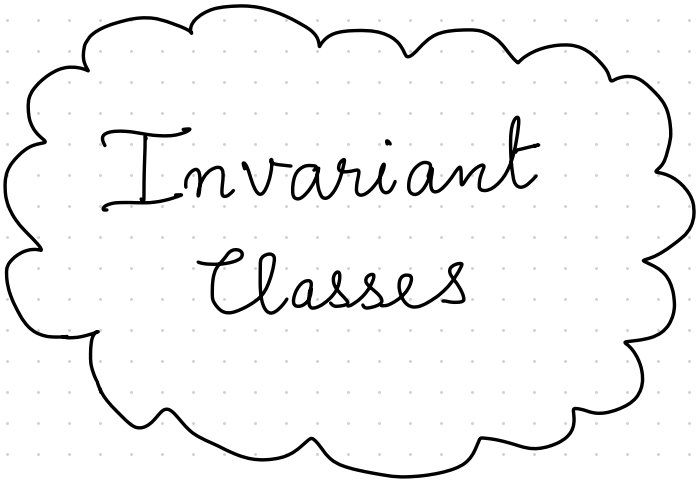
Anomalies for global symmetries are not necessarily problematic, however, anomalies for BRST symmetries are because then the BRST transformation of the effective action will produce negative norm ghosts!



Just your
friendly
neighbour ghost.

Gauge Anomalies should really be called BRST anomalies.

Before we jump to anomalies, let's take a quick side tour of invariant classes.



Invariant Classes

Invariant Classes

Invariant classes are subsets of de-Rham Cohomology class. They are invariant under gauge (BRST) transformations and entail important topological properties about the bundle.

Examples: $F = dA + \frac{1}{2} [A, A]$

1.) Chern Class:

$$C(F) = \det(1 + F) ;$$

$$C_0 = 1, C_1 = \text{tr}(F), C_2 = \frac{1}{2} (\text{tr}(F)^2 - \text{tr}(F \wedge F))$$

$$C_K = \det(F)$$

terminates at $K = \dim(g)$; $C_n \in H^{2n}(M, g)$

2.) Chir Character :

$$\text{Ch}(F) = \text{Tr}(\exp(F))$$

$$\text{Ch}_0 = \dim(g) ; \text{Ch}_1 = \text{Tr}(F) = C_1$$

$$\text{Ch}_2 = \frac{1}{2} \text{Tr}(F \wedge F) ; \dots \text{Ch}_n = \frac{1}{n!} \text{Tr}[F \wedge \dots \wedge F] \\ \in H^{2n}(M, g).$$

A useful property of Chern Class and Chern characters is that $C(E \otimes F) = C(E) \wedge C(F)$
and $Ch(E \otimes F) = Ch(E) \wedge Ch(F)$ for bundles E & F .

$$3) \text{ Todd Class: } Td(F) = \prod_i \frac{x_i}{1 - e^{x_i}}$$

where x_i are the Chern-roots (i.e., eigen values of F)

$$Td_0 = 1 ; Td_1 = \frac{1}{2} C_1 ; Td_2 = \frac{1}{12} (C_1^2 + C_2)$$

$$Td_3 = \frac{1}{24} C_1 C_2 ; Td_4 = \frac{1}{720} (-C_1^4 + 4C_1^2 C_2 + 3C_2^2 + 4C_3 - C_4)$$

Newton's Identities

$$Ch_1 = C_1$$

$$Ch_2 = \frac{1}{2} (C_1^2 - 2C_2)$$

$$Ch_3 = \frac{1}{6} (C_1^3 - 3C_2 C_1 + 3C_3)$$

$$Ch_4 = \frac{1}{24} (C_1^4 - 4C_2 C_1^2 + 4C_3 C_1 + 2C_2^2 - 4C_4)$$

⋮

That is all we need to know for anomaly cancellation on twistor space!



Anomalies

Chiral Anomaly v/s Gauge Anomaly

We have already seen that anomaly is the variation of the effective action under symmetry.

The classically conserved currents are no longer conserved quantum mechanically.

The action $S = \int_{\mathbb{R}^4} d\mu \bar{\Psi} \gamma^\mu i(\partial_\mu + A_\mu) \Psi$

is invariant classically under the global $U(1)$ symmetry

$$\Psi \rightarrow e^{i\alpha\gamma_5} \Psi$$

The current is $j_5^\mu = \bar{\Psi} \gamma^5 \gamma^\mu \Psi$

The anomaly is the non-conservation of the current

$$\partial_\mu j_5^\mu = \frac{1}{16\pi^2} \epsilon^{\alpha\beta\gamma\delta} \text{tr}(F_{\alpha\beta} F_{\gamma\delta})$$

$$= \frac{1}{16\pi^2} \epsilon^{\alpha\beta\gamma\delta} \partial_\alpha \text{tr} \left(A_\beta \partial_\gamma A_\delta + \frac{2}{3} A_\beta A_\gamma A_\delta \right)$$

$\underbrace{\hspace{10em}}_{CS_3 \rightarrow \text{3D Chern Simons form.}}$

Consider $S = \int_{\mathbb{R}^4} d\mu \psi^\dagger \gamma^\mu (\nabla_\mu) P_+ \psi$; $P_+ = \frac{1+\gamma_5}{2}$

It is invariant under the gauge transformations:

$$A \rightarrow g(d+A)g^{-1} \text{ and } \psi \rightarrow g\psi$$

The corresponding current is Lie algebra valued

$$j^{\mu a} = \psi^\dagger \gamma^\mu T^a P_+ \psi$$

The gauge anomaly is the non-conservation of this current.

$$D_\mu j^{\mu a} = \frac{1}{24\pi^2} \epsilon^{\alpha\beta\gamma\delta} \partial_\alpha \text{tr} \left[A_\beta \partial_\gamma A_\delta + \frac{1}{2} A_\beta A_\gamma A_\delta \right]$$

↓
This is ~~not~~ the 3D Chern
Simons form

Note the $1/2$ instead of $2/3$ for gauge anomalies.

This is very important and we will see that gauge anomalies in $2m$ dimensions are related to chiral anomalies in $2m+2$ dimensions by something called descent relations.

Chiral (Abelian) anomalies (Fujikawa)

$$Z = \int d\bar{\psi} d\psi e^{-S}$$

$$\begin{aligned} \delta Z &= \int \delta[d\bar{\psi} d\psi] e^{-S} + [d\bar{\psi} d\psi] e^{-\delta S} \\ \downarrow \\ \text{Chiral} \\ U(1) &= \int \delta[d\bar{\psi} d\psi] e^{-S} + [d\bar{\psi} d\psi] e^{-S - \int \partial_\mu j_5^\mu} \end{aligned}$$

$$\text{Choose a basis } \psi = \sum_i a_i \psi_i ; \quad \bar{\psi} = \sum_i b_i \psi_i^\dagger$$

a_i and b_i are Grassmann numbers.

$$\langle \psi_i | \psi_j \rangle = \int d^4x \psi_i^\dagger \psi_j = \delta_{ij} \quad (\text{orthonormal})$$

Under Chiral transformation

$$\begin{aligned} \psi' &= \psi + i\alpha \gamma_5 \psi = \sum_i a'_i \psi_i \\ \bar{\psi}' &= \bar{\psi} + i\bar{\psi} \alpha \gamma_5 = \sum_i b'_i \psi_i^\dagger \end{aligned}$$

The path integral measure

$$\prod_i a_i b_i \rightarrow \prod_i a'_i b'_i$$

$$\begin{aligned} a'_i &= \langle \psi_i | \psi' \rangle = \langle \psi_i | 1 + i\alpha \gamma_5 | \psi \rangle \\ &= \sum_j \langle \psi_i | 1 + i\alpha \gamma_5 | \psi_j \rangle a_j = \sum_j c_{ij} a_j \end{aligned}$$

$$C_{ij} = \delta_{ij} + \langle \psi_i | i \alpha \gamma_5 | \psi_j \rangle$$

$$\begin{aligned} \Rightarrow \prod_i da_i &= \det(C_{ij})^{-1} \prod da_i \\ &= \exp(-\text{tr} \ln(\delta_{ij} + \langle \psi_i | i \alpha \gamma_5 | \psi_j \rangle)) \prod da_i \\ &\approx \exp(-\text{tr}(\langle \psi_i | i \alpha \gamma_5 | \psi_j \rangle)) \prod da_i \\ &= \exp(-\sum_i \langle \psi_i | i \alpha \gamma_5 | \psi_i \rangle) \prod da_i \end{aligned}$$

$$\text{Similarly for } \prod db_i \rightarrow \exp(-\sum_i \langle \psi_i | i \alpha \gamma_5 | \psi_i \rangle) \prod db_i$$

$$\therefore \delta Z = \int \left[\prod_i da_i db_i \right] e^{-S - \int \partial_\mu j_5^\mu - 2i\alpha \sum_i \langle \psi_i | \gamma_5 | \psi_i \rangle}$$

$$\sum_i \langle \psi_i | \gamma_5 | \psi_i \rangle = \sum_i \int \psi_i^\dagger(x) \gamma_5 \psi_i(x) d^4x = \int A(x) d^4x$$

$$\Rightarrow \partial_\mu j_5^\mu = -2i\alpha A(x)$$

where $A(x) = \sum_i \psi_i^\dagger(x) \gamma_5 \psi_i(x)$ is the chiral anomaly.

Further assume that our orthonormal basis Ψ_i s are also eigen-states of $i\not{D}$. It is always possible to choose such a basis.

$$i\not{D} \Psi_i = \lambda_i \Psi_i$$

$$\begin{aligned} \int dx A(x) &= \sum_i \langle \Psi_i | \gamma_5 | \Psi_i \rangle \\ &= \lim_{M \rightarrow \infty} \sum_i \langle \Psi_i | \gamma_5 e^{-(i\not{D}/M)^2} | \Psi_i \rangle \end{aligned}$$

Here $M \rightarrow \infty$ is the same as $t = \frac{1}{M} \rightarrow 0$

This is the heat-kernel regulator.

If $|\Psi_i\rangle$ has eigen value λ_i then $\gamma_5 |\Psi_i\rangle$ has eigen value $-\lambda_i$ since $\{\gamma_5, i\not{D}\} = 0$.

$\therefore \langle \Psi_i | \gamma_5$ will be orthogonal to $|\Psi_i\rangle$ unless $\lambda_i = 0$

$\therefore$ Only the zero eigen modes will contribute to the anomaly.

$$\begin{aligned} \Rightarrow \int dx A(x) &= \sum_i \left[\langle 0_+, i | 0_+, i \rangle - \langle 0_-, i | 0_-, i \rangle \right] \begin{matrix} \nearrow 0 \text{ eigen states} \\ \text{with (+)ve chirality} \end{matrix} \begin{matrix} \nearrow 0 \text{ eigen-states} \\ \text{with (-)ve} \\ \text{chirality} \end{matrix} \\ &= \nu_+ - \nu_- = \text{ind}(i\not{D}) \end{aligned}$$

$\nu_{\pm} \rightarrow$ no. of left (right) handed spinors with zero eigen value

The Chiral anomaly is the index of the Dirac operator

Atiyah-Singer Index Theorem

Good friends $\longrightarrow$



The index of the Dirac operator in $2m$ dimension is given by the m^{th} Chern character of its curvature.

$$D = d + A, \quad (d + A)^2 = F$$

$$\nu_+ - \nu_- = \int_M \text{Ch}_m(F) = \int_M \frac{1}{m!} \text{Tr} [F \wedge \dots \wedge F]$$

$\downarrow$
m-times

We will give a proof in 4D.

$$\begin{aligned} (i \not{D})^2 &= -\gamma^\mu \gamma^\nu \nabla_\mu \nabla_\nu \\ &= -\left(\gamma^{\mu\nu} + \frac{1}{2} [\gamma^\mu, \gamma^\nu]\right) \left(\frac{1}{2} \{\nabla_\mu, \nabla_\nu\} + \frac{1}{2} F_{\mu\nu}\right) \\ &= -\nabla_\mu \nabla^\mu - \frac{1}{4} [\gamma^\mu, \gamma^\nu] F_{\mu\nu} \end{aligned}$$

$$A(x) =$$

$$\sum_i \langle \psi_i | x \rangle \left[\gamma_5 e^{-(\nabla_\mu \nabla^\mu + \frac{1}{4} [\gamma^\mu, \gamma^\nu] F_{\mu\nu}) / M^2} \right] \langle x | \psi_i \rangle \Big|_{M \rightarrow \infty}$$

$$\Rightarrow A(x) = \int \frac{d^4 k}{(2\pi)^4} \frac{d^4 k'}{(2\pi)^4}$$

$$\sum_i \langle \psi_i | k' \rangle \langle k' | x \rangle \left[\gamma_5 \exp \left[-(\nabla_\mu \nabla^\mu + \frac{1}{4} [\gamma^\mu, \gamma^\nu] F_{\mu\nu}) / M^2 \right] \right. \\ \left. \langle x | k \rangle \langle k | \psi_i \rangle \right] \Big|_{M \rightarrow \infty}$$

Using the completeness of the basis

$$\sum_i \langle \psi_i | k' \rangle \langle k | \psi_i \rangle = (2\pi)^4 \delta^4(k' - k) \text{ and integrating } k'$$

$$\Rightarrow A(x) = \int \frac{d^4 k}{(2\pi)^4} \text{tr} \left[\gamma_5 \exp \left(\frac{-k^2}{M^2} - \frac{[\gamma^\mu, \gamma^\nu] F_{\mu\nu}}{4 M^2} \right) \right] \Big|_{M \rightarrow \infty}$$

define $\tilde{k} = k/M$

$$A(x) = \text{tr} \left[\gamma_5 \exp \left(\frac{[\gamma^\mu, \gamma^\nu] F_{\mu\nu}}{4 M^2} \right) \right] M^4 \underbrace{\int \frac{d^4 \tilde{k}}{(2\pi)^4} e^{-\tilde{k}^2}}_{\frac{\pi^2}{(2\pi)^4}} \Big|_{M \rightarrow \infty}$$

$$\text{tr}(\gamma_5) = \text{tr}(\gamma_5 \gamma_\mu \gamma_\nu) = 0$$

$$\text{tr}(\gamma_5 \gamma_\mu \gamma_\nu \gamma_\rho \gamma_\sigma) = -4 \epsilon_{\mu\nu\rho\sigma}$$

$$\Rightarrow A(x) = \frac{1}{16\pi^2} \text{tr} \left[\frac{\gamma_5}{2!} \left(\frac{[\gamma^\mu, \gamma^\nu] F_{\mu\nu}}{4} \right)^2 \right] \\ = -\frac{1}{32\pi^2} \epsilon^{\alpha\beta\gamma\delta} \text{tr}(F_{\alpha\beta} F_{\gamma\delta})$$

$$\boxed{d^4 x \partial_\mu j_5^\mu = -\frac{1}{16\pi^2} F \wedge F} \rightarrow \text{Chiral Anomaly!} \\ \text{in 4D.}$$

Gauge Anomalies

We had promised to show that gauge anomalies in $2m$ dimensions are related to chiral anomalies in $2m+2$ dimensions which are given by the m -th Chern character.

Gauge Anomaly is the failure of the effective action to be invariant under BRST transformation.

$$e^W = Z = \int [D\bar{\Psi} D\Psi DA] e^{-S}$$

$$G := \delta_B W = \text{Gauge Anomaly}$$

$\therefore$ the gauge anomaly is BRST exact

$$\Rightarrow \boxed{\delta_B G = 0}$$

This is known as the Wess-Zumino Consistency condition.

(Wess and Zumino don't have a picture together)

So, if we want to write an expression for gauge anomaly it must be BRST exact.

Suppose Q_{2m-1} is the Chern-Simons form for the m th Chern Character (which is a $2m$ -form)

$$\Rightarrow dQ_{2m-1}(A, F) = \text{Ch}_m(F)$$

but $\delta_B \text{Ch}_m(F) = 0$ (this is the invariance in invariant class)

$$\Rightarrow \delta_B dQ_{2m-1} = 0$$

$$\Rightarrow -d\delta_B Q_{2m-1} = 0$$

$$\Rightarrow \delta_B Q_{2m-1} \text{ is closed}$$

In a local patch any closed form is exact

$$\Rightarrow \delta_B Q_{2m-1}(A, F) \underset{\substack{\downarrow \\ \text{local patch}}}{=} dP_{2m-2}^1(c, A, F)$$

Here P_{2m-2}^1 is just a $(2m-2)$ form which also depends on the ghost field c since it is related to the BRST variation of Q_{2m-1} . The label 1 in P_{2m-2}^1 denotes that it is linear in c .

These relations are referred to as "descent relations".

There is a more formal development of this that we will avoid for now.

$$\therefore \text{Ch}_m \rightarrow dQ_{2m-1} \quad \text{and} \quad \delta_B Q_{2m-1} \rightarrow dP_{2m-2}^1$$

The Gauge Anomaly has to be : i) topological
ii) BRST exact

$\therefore P_{2m-2}^1$ is the perfect candidate.

Example 1: Lets calculate P_2^1

We start with $\text{Ch}_2 = \text{tr}[F \wedge F] = d \text{tr} \left(A dA + \frac{2}{3} A \wedge A \wedge A \right)$

We get $Q_3 = \text{tr} \left[A dA + \frac{2}{3} A \wedge A \wedge A \right] = \text{tr} \left[AF - \frac{1}{3} A \wedge A \wedge A \right]$

Lets calculate the BRST transformation of Q_3 .

$$\delta_B A = dA + [A, c] ; \quad \delta_B F = [F, c]$$

$$\delta_B A^l = dc A^{l-1} + A dc A^{l-2} + \dots + A^{l-1} dc + [A^l, c]$$

$$\delta_B \text{tr}(A^l F^k) = \text{tr}(dc A^{l-1} + A dc A^{l-2} + \dots + A^{l-1} dc) F^k$$

$$(\because \text{tr}[A^l F^k, c] = 0)$$

$$\begin{aligned}
\therefore \delta_B Q_3 &= \delta_B \operatorname{tr} \left[AF - \frac{1}{3} A^3 \right] \\
&= \operatorname{tr} [dc F - dc A^2] \\
&= \operatorname{tr} [dc \wedge dA] \\
&= d \operatorname{tr} [c \wedge dA]
\end{aligned}$$

$$\therefore P_2^1 = c dA$$

Example 2 P_4^1

$$\begin{aligned}
Ch_3 &\propto \operatorname{tr} [F \wedge F \wedge F] \\
&= d \operatorname{tr} \left[A dA \wedge dA + \frac{3}{2} A^3 dA + \frac{3}{5} A^5 \right] \\
&= d \operatorname{tr} \left[AF^2 - \frac{1}{2} A^3 F + \frac{1}{10} A^5 \right]
\end{aligned}$$

$$\begin{aligned}
\delta_B Q_5 &= \delta_B \operatorname{tr} \left[AF^2 - \frac{1}{2} A^3 F + \frac{1}{10} A^5 \right] \\
&= \operatorname{tr} \left[dc F^2 - \frac{dc}{2} [A^2 F + AFA + FA^2] + \frac{1}{2} dc A^4 \right]
\end{aligned}$$

All the $dc A^4$ terms cancel

$$\begin{aligned}
\delta_B Q_5 &= \operatorname{tr} \left[dc \left[dA dA + \frac{1}{2} (dA A^2 - A dA A + A^2 dA) \right] \right] \\
&= \operatorname{tr} \left[dc \left[dA dA + \frac{1}{2} d(A^3) \right] \right]
\end{aligned}$$

$$\Rightarrow P'_4 = \text{tr} \left[c d \left[A dA + \frac{1}{2} A^3 \right] \right]$$

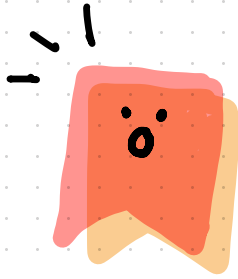
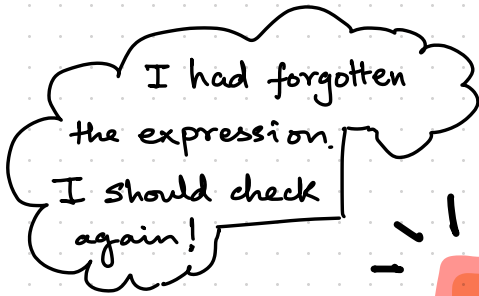
Notice that the gauge anomaly in 4D was given by

$$\partial_\mu j^\mu \propto \epsilon^{\mu\nu\rho\sigma} \text{tr} \left[\partial_\mu \left[A_\nu \partial_\rho A_\sigma + \frac{1}{2} A_\nu A_\rho A_\sigma \right] \right]$$

↓

Remember

that chiral anomaly
had $\frac{2}{3}$ making it a
Chern Simons form.



Gauge anomaly is really BRST anomaly.

Presence of Gravity

To calculate the gauge anomaly in $2m$ dimension
write $m+1$ dimensional Chern Character C_{m+1}
and its Chern Simons form Q_{2m+1} .

The gauge anomaly is P'_{2m} such that $dP_{2m}^1 = \delta_B Q_{2m+1}$

This algorithm works in flat space.

But there is a much deeper reason for this which is visible if we are in curved space.

If we have a bundle E over a manifold M the Euler characteristic of the bundle is given by

$$\chi(E) = \int Td(T_M) \wedge Ch(E) \quad (\text{Hirzebruch Riemann-Roch Theorem})$$

T_M is the tangent bundle of M .

For flat space this reduces to $\int Ch(E)$

but in general we should extract the $2m+2$ form from

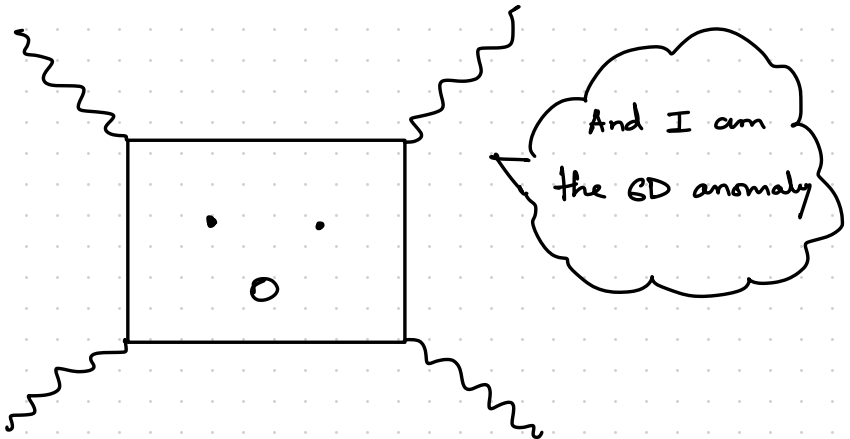
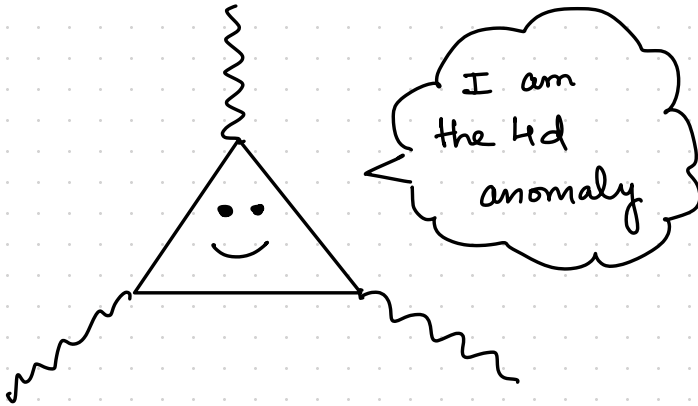
$$\left(\int_M Td(T_M) \wedge Ch(E) \right)_{2m+2}$$

and get the corresponding $2m$ form by descent relation to get the gauge, gravitational, and mixed gauge-gravitational anomalies.

Physically the anomalies will correspond to the $m+1$ loop diagrams as can be seen from the interaction term in the anomaly.

Q: How does it make sense to integrate a $2m+2$ form on M which is $2m$ dimensional?

Answer: It doesn't. This is a trick to get the anomaly polynomial. Think of joining a 2D disk at each point of the manifold to make a $2m+2$ dimensional manifold. The integral then makes sense.



Anomalies on Twistor Space

Let's go back to twistor theory.

Remember that in SD Einstein gravity we have the fibration $\mathbb{P}\mathbb{D}^3 \rightarrow \mathbb{CP}^1$

which is called a Hamiltonian fibration since the Poisson bracket along the fibres is preserved.

Due to this fibration things are easier to calculate in SD Einstein gravity than they are in SD conformal gravity.

Let's calculate the anomalies in SD Einstein first.

Positive helicity graviton is an $\mathcal{O}(2)$ bundle over $\mathbb{CP}^1$

Negative helicity graviton is an $\mathcal{O}(-6)$ bundle over $\mathbb{CP}^1$.

- Using the Master formula, the anomaly for pure gravity is the $(4,4)$ form extracted from

$$\frac{1}{2} \int_{\mathbb{P}\mathbb{D}^3} \text{Td}(T_{\mathbb{P}\mathbb{D}^3}) \wedge [\text{Ch}(\mathcal{O}(2)) + \text{Ch}(\mathcal{O}(-6))]$$

Similarly, positive helicity gauge bosons are Lie algebra valued and $\mathcal{O}(0)$ bundles over $\mathbb{P}^1$

whereas, negative helicity gauge bosons are Lie algebra valued and $\mathcal{O}(-4)$ bundles over $\mathbb{P}^1$.

The anomaly polynomial is

$$\left[\frac{1}{2} \int_{\mathbb{P}^3} \text{Td}(T_{\mathbb{P}^3}) \wedge \text{Ch}(g) \wedge [\text{Ch}(\mathcal{O}(-4)) + 1] \right]_{(4,4)}.$$

Let's simplify this.

$$\begin{aligned} \text{Ch}(g) &= \text{Ch}_0(g) + \text{Ch}_2(g) + \text{Ch}_4(g) + \dots \quad \rightarrow \text{not useful terms} \\ &= \dim(g) + \text{Ch}_2(g) + \text{Ch}_4(4) \end{aligned}$$

(The odd terms vanish in the adjoint representation)

The Chern character of any line bundle over $\mathbb{P}^1$ is given by

$$\text{Ch}(\mathcal{O}(n)) = 1 + C_1(\mathcal{O}(n)) = 1 - (n/2)x$$

line bundles are 1-dimensional

where x is the Fubini Study metric on $\mathbb{CP}^1$

$$x = \frac{dz \wedge d\bar{z}}{(1+|z|^2)^2}$$

Since this is the top form on $\mathbb{CP}^1$ $x^2 = 0$.

Remember that in SD Einstein gravity

$$K_{\mathbb{CP}^1} \cong \mathcal{O}(-4) \therefore c_1(K) = 2x$$

↳ Canonical
bundle (top-form)

Canonical bundle in general over any complex manifold is line bundle.

$$c(K) = c(\det(T)^{-1}) = 1 + c_1(T^*) = 1 - c_1(T)$$

In our case, we therefore obtain that $c_1(T) = -2x$.

$$\begin{aligned} Td(T) &= 1 + \overset{\mapsto Td_0}{\frac{c_1(T)}{2}} + \overset{\mapsto Td_1}{\frac{1}{12}(c_1^2 + c_2)} + \overset{\mapsto Td_2}{\frac{1}{24}c_2} \\ &\quad + \overset{\mapsto Td_3}{\frac{1}{720}(-c_1^4 + 4c_1^2c_2 + 3c_2^2 + c_3 - c_4)} \\ &\quad + \dots \mapsto \text{not useful terms} \end{aligned}$$

If we write c_n or Ch_n henceforth it means $c_n(T)$, $Ch_n(T)$.

Using the simplification from $C_1^2 = 0$, this becomes

$$Td(T) = 1 + \frac{C_1}{2} - \frac{1}{12} Ch_2 - \frac{1}{24} C_1 Ch_2 + \frac{1}{720} (3C_2^2 + C_1 C_3)$$

we have dropped $C_4(T)$ since $\dim(T) = 3$
and Chern class terminates at C_3 .

To calculate $C_3(T)$ we can break T into
the $T_{\mathbb{CP}^1}$ base space and T_N along the fibres.

$$C(T_{\mathbb{CP}^1} \wedge T_N) = C(T_{\mathbb{CP}^1}) \wedge C(T_N)$$

$$\Rightarrow C_3(T_{\text{proj}}) = C_3(T_N) + C_1(T_{\mathbb{CP}^1}) C_2(T_N)$$

$$C_3(T_N) = 0 \quad \because \dim(T_N) = 2; \quad C_1(T_{\mathbb{CP}^1}) = x$$

$$\Rightarrow C_3(T_{\text{proj}}) = x C_2(T_N)$$

In our expression above for Todd class the $C_1 C_3$
therefore becomes $C_1^2 C_2(T_N) = 0$.

$$\therefore Td(T) = 1 + \frac{C_1}{2} - \frac{1}{12} Ch_2 - \frac{1}{24} C_1 Ch_2 + \frac{1}{720} (3C_2^2)$$

Using Newton's identities;

$$Ch_4 = \frac{1}{4!} (C_1^4 - 4C_2 C_1^2 + 4C_3 C_1 + 2C_2^2 - 4C_4) = C_2^2 / 12$$

$$\Rightarrow \frac{C_2^2}{240} = \frac{Ch_4}{20}$$

Getting
There

$$\therefore Td(T) = 1 + \frac{C_1}{2} - \frac{1}{12} Ch_2 - \frac{1}{24} C_1 Ch_2 + \frac{1}{20} Ch_4$$

$$(1 + Ch(\theta(-4))) = 1 + 1 - \left(\frac{-4}{2}\right)x = 2 + 2x = 2 - C_1$$

$$\begin{aligned} (Ch(\theta(2)) + Ch(\theta(-6))) &= 1 - (2/2)x + 1 - (-6/2)x \\ &= 2 - 2x = 2 + C_1 \end{aligned}$$

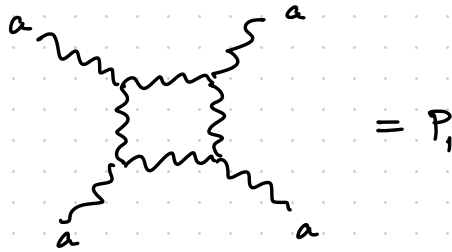
$$\Rightarrow \left[\frac{1}{2} Td(T) (Ch(\theta(2)) + Ch(\theta(-6))) \right]_{(4,4)} = \frac{Ch_4}{20}$$

$$\begin{aligned} \text{and } \left[\frac{1}{2} Td(T) \wedge Ch(g) \wedge (Ch(\theta(-4)) + 1) \right]_4 \\ = Ch_4(g) - \frac{1}{12} Ch_2(g) Ch_2(T) + \frac{\dim(g)}{20} Ch_4(T) \end{aligned}$$

$\Rightarrow$ The total anomaly polynomial is:

$$\boxed{\int Ch_4(g) + \frac{\dim(g)+1}{20} \int Ch_4(T) - \frac{1}{12} \int Ch_2(g) Ch_2(T)}$$

- $\int \text{Ch}_4(g)$ gives us the following diagram



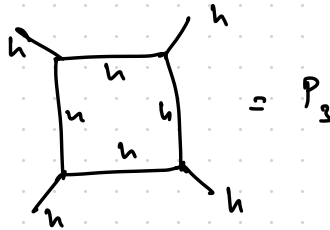
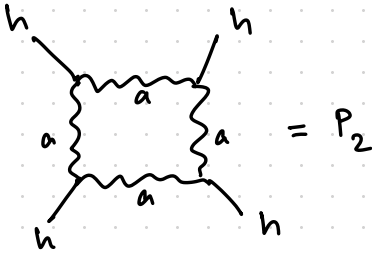
This is the gauge anomaly on twistor space.

By descent it is given by

$$P_1 = \int_{\mathbb{P}^3} \text{Tr}[c(\partial a)^3];$$

$$\begin{matrix} \nearrow \text{BRST ghost} \\ c = g \delta_B g^{-1} \end{matrix}$$

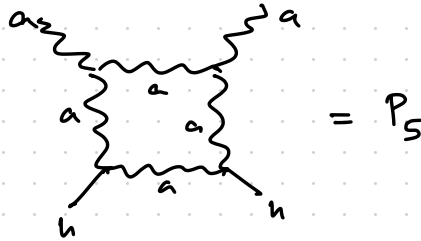
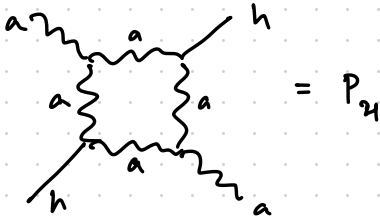
- $\frac{\dim(g)+1}{20} \int \text{Ch}_4(T)$ gives two possible diagrams



$$P_2 = \dim(g) \int \text{Tr}[u(\partial s)^3]; \quad P_3 = \int \text{Tr}[u(\partial s)^3]$$

s is the $sl(2, \mathbb{C})$ connection on $T_{\mathbb{P}^3}$ encountered before and u is its BRST ghost.

- $-\frac{1}{12} \int \text{Ch}_2(g) \text{Ch}_2(T)$ gives two diagrams



This is the gauge-gravitational mixed anomalies.

Fermionic Anomalies

$$S_F = \int_{\mathbb{P}^3} \bar{\Psi} \not{D} \Psi$$

$$\Psi \in \Omega^{0,1}(\mathbb{P}^3, \mathcal{O}(-1))$$

$$\bar{\Psi} \in \Omega^{3,1}(\mathbb{P}^3, \mathcal{O}(1))$$

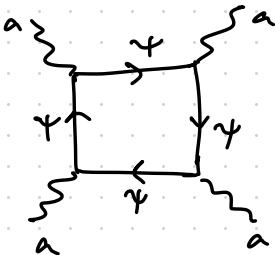
$$\cong \Omega^{0,1}(\mathbb{P}^3, \mathcal{O}(-3))$$

The Anomaly is

$$\int -\frac{1}{2} \text{Td}(T) \wedge [\text{Ch}(V) \wedge \text{Ch}(\mathcal{O}(-1)) + \text{Ch}(V^*) \wedge \text{Ch}(\mathcal{O}(-3))]$$

Fermions
are very negative
beings!

V is some representation of the
gauge group G and V^* is its
dual representation.

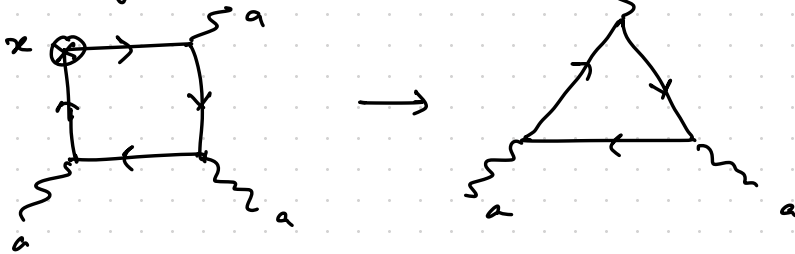


In Euclidean signature

$$\mathbb{PT} \cong \mathbb{R}^4 \times \mathbb{CP}^1$$

So going from $\mathbb{PT}$ to $\mathbb{R}^4$ corresponds to compactifying the $\mathbb{CP}^1$.

The diagrams then reduce to



These are the well known triangle diagrams in spacetime.

Conformal Gravity

The conformal gravity case is more complicated since we no longer have the fibration

$$\mathbb{P}\mathbb{Y} \rightarrow \mathbb{CP}^1.$$

In this case, the gravitons are given by

$$g \in \Omega^{0,1}(\mathbb{P}\mathbb{Y}, T_{\mathbb{P}\mathbb{T}})$$

$$h \in \Omega^{3,1}(\mathbb{P}\mathbb{Y}, T_{\mathbb{P}\mathbb{T}}^*) \cong \Omega^{0,1}(\mathbb{P}\mathbb{Y}, T_{\mathbb{P}\mathbb{T}}^* \otimes K)$$

$\therefore$ The anomaly polynomial for gravity is given by

$$\frac{1}{2} \text{Td}(T) \wedge [\text{Ch}(T) + \text{Ch}(T^* \otimes K)]$$

$$\begin{aligned} \text{Ch}(T) &= \overset{\mapsto \dim(T)}{3} + \overset{\mapsto \text{Ch}_1(T)}{c_1(T)} + \overset{\mapsto \text{Ch}_2}{\frac{1}{2}(c_1^2 - 2c_2)} + \overset{\mapsto \text{Ch}_3}{\frac{1}{6}(c_1^3 - 3c_2c_1 + 3c_3)} \\ &\quad + \frac{1}{24}(c_1^4 - 4c_2c_1^2 + 4c_3c_1 + 2c_2^2 - 4c_4) \\ &\quad \mapsto \text{Ch}_4 \end{aligned}$$

$$\text{Ch}(T^*) = 3 - \text{Ch}_1(T) + \text{Ch}_2(T) - \text{Ch}_3(T) + \text{Ch}_4(T)$$

$$\text{Note: } c_4 = 0 \quad \because \dim(T) = 3.$$

K is the Canonical bundle and it is true that $C(K) = 1 - C_1(T)$

$$\Rightarrow \text{Ch}(K) = \underset{\downarrow \text{dim}(K)}{1} - \underset{\downarrow \text{Ch}_1(K)}{C_1} + \underset{\downarrow \text{Ch}_2(K)}{\frac{1}{2} C_1^2} - \underset{\downarrow \text{Ch}_3(K)}{\frac{1}{6} C_1^3} + \underset{\downarrow \text{Ch}_4(K)}{\frac{1}{24} C_1^4} + \dots$$

∴ Upon calculation one gets

$$\left[\frac{1}{2} Td(T) \wedge [\text{Ch}(T) + \text{Ch}(T^*) \wedge \text{Ch}(K)] \right]_{(4,4)} = P$$

$$P_0 = \frac{1}{240} \left[39 C_1^4 - 96 C_1^2 C_2 + 101 C_1 C_3 + 3 C_2^2 \right]$$

Note that in the SD Einstein case, we only get

$$\frac{1}{20} \int \text{Ch}_4 = \frac{1}{240} \int C_2^2$$

Here we have extra terms C_1^4 , $C_1^2 C_2$, and $C_1 C_3$

On reducing to spacetime by compactifying $\mathbb{CP}^1$

these will become linear combinations of the 'a' and 'c' anomalies.

For gauge anomaly we have:

$$\frac{1}{2} \int Td(T) \wedge \text{Ch}(g) [1 + \text{Ch}(\mathcal{O}(-4))]$$

We do not have fibrations over $\mathbb{C}P^1$ but
we can replace $\text{Ch}(\mathcal{O}(n))$ with $\text{Ch}(\mathcal{O}(nK^{-n/4}))$

We get

- Pure gravity: $\frac{1}{240} [39c_4^4 - 96c_4^2c_2 + 1014c_3 + 3c_2^2] = P_0$
in absence
of gauge bosons
- Pure gauge: $\text{Ch}_4(g) = P_1$
- Pure gravity: $P_0 + \dim(g) Td_4$
 $= P_0 + \frac{1}{720} (-c_4^4 + 4c_4^2c_2 + 3c_2^2 + 4c_3)$
- Mixed gauge gravity: $\frac{1}{12} \text{Ch}_2(g) [c_1^2 + c_2]$

Anomaly
Cancellation

Anomaly Cancellation (SD Einstein case)

To cancel the gauge anomaly Costello introduces the η field on twistor space. $\eta \in \Omega^{2,1}(\mathbb{P}^1)$ such that $\bar{\partial}\eta = 0$. η is closed. The action of this field is

$$\int_{\mathbb{P}^1} (\bar{\partial}\eta)(\bar{\partial}^1\eta)$$

if $\bar{\partial}\eta = \chi$ then $\eta = \partial\chi$.

η resembles the closed string modes in type IIB topological strings but more importantly the action for this field on spacetime is

$$\int_{\mathbb{R}^4} \phi \square^2 \phi \quad \text{where } \phi \text{ is a dimension 0 scalar.}$$

To cancel the gauge anomaly we should couple it to the gauge bosons

$$S = \int (\bar{\partial}\eta)(\bar{\partial}^1\eta) + \lambda_g \eta \text{tr}(a\partial a)$$

The BRST transformations for BF are now given by

$$\delta \eta = \bar{\partial} \theta - \lambda_g \operatorname{tr}(\partial c \partial a)$$

$$\delta a = \bar{\partial} c + [a, c]$$

$$\delta b = -[b, c] + \bar{\partial} d + [a, d] - i\lambda_g(\theta \partial a + \eta \partial c)$$

c and d are the standard ghost fields

$$\theta \in \Omega^{2,0} \text{ st. } \partial \theta = 0.$$

The action $S = \int \bar{\partial}^\dagger \eta \bar{\partial} \eta + \lambda_g \eta \operatorname{tr}(a \partial a)$

is manifestly not invariant under BRST

$$\delta S = -\lambda_g^2 \int \operatorname{tr}(c \partial a) \operatorname{tr}((\partial a)^2)$$

Recall that The anomaly was

$$\int \operatorname{tr}[c(\partial a)^3]$$

We can use the classical invariance of the η action to cancel the quantum anomaly (this is the Green Schwarz mechanism)

This will be true if $(\operatorname{tr}(x^2))^2 \propto \operatorname{tr}(x^4)$

x being the generator of $\mathfrak{g}$.

This is true only if $\mathfrak{g} = \mathfrak{sl}(2, \mathbb{C}), \mathfrak{sl}(3, \mathbb{C}), \mathfrak{so}(8),$
 $\mathfrak{g}_2, f_4, e_6, e_7, e_8.$

for which

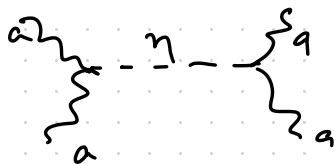
$$\text{tr}(\chi^4) = \frac{5(2h^\vee)^2}{2(2 + \dim(\mathfrak{g}))} (\text{tr}(\chi^2))^2$$

$h^\vee$ is the dual Coxeter number of $\mathfrak{g}.$

The coupling λ_g^2 should be carefully chosen to cancel the anomaly

$$\lambda_g^2 = \frac{C_g}{3!} \left(\frac{i}{2\pi} \right)^2$$

The classical BRST transformation of the action corresponds to this diagram:



Similarly to cancel the gravitational anomaly we need an action

$$S_\eta = \int \bar{\partial} \eta \bar{\nabla} \eta + \mu \eta \text{tr}[s \partial s]$$

$$\bar{\nabla} \eta = \bar{\partial} \eta + \{h, \cdot\} \lrcorner \eta$$

The BRST transformation of this action is

$$\delta h = \bar{\partial} \chi + \{h, \chi\}$$

$$\delta \eta = \bar{\nabla} \Theta - \mu \text{tr}(\partial \psi \partial s)$$

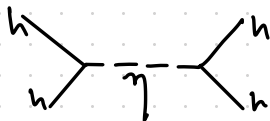
$$\begin{aligned} \delta g = & -\{g, \chi\} + \bar{\partial} \phi + \{h, \phi\} \\ & + \frac{i}{2} \partial_{\alpha} (\Theta \partial^{\alpha} \lrcorner \eta + \eta \partial^{\alpha} \lrcorner \Theta) \\ & + i\mu ((\partial^{\beta} \partial_{\alpha} \eta)(\partial \psi^{\alpha}_{\beta}) + (\partial^{\beta} \partial_{\alpha} \Theta)(\partial s^{\alpha}_{\beta})) \end{aligned}$$

h transforms
same as before
but g 's transformation
is monstrous



The BRST variation of the action then gives

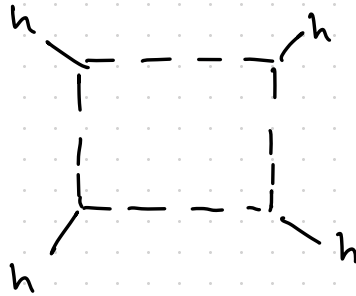
$$\delta S \propto \mu^2 \int \text{tr}(\psi \partial s) \text{tr}(\partial s)^2$$



This again cancels the anomaly

Since $\text{tr}(X^4) \propto (\text{tr}(X^2))^2$ is true for $SL(2, \mathbb{C})$ connections.

One needs to be careful this time since the graviton couples with the kinetic term in the η lagrangian and η contributes to the anomaly by going in the loop.



The contribution is the same as when we have gravitons in the loop and this is given by

$$\int \text{tr}(\eta (\partial s)^3)$$

Therefore the gravitational anomaly becomes twice as before,

μ^2 should carefully be chosen to be $\frac{1}{5!} \left(\frac{i}{2\pi}\right)^2$

to cancel the anomaly

The η field can simultaneously cancel gauge, gravitational, and gauge-gravitational anomalies if

$$\lambda_g^2 = \frac{C_g}{3!} \left(\frac{i}{2\pi} \right)^2 \quad \text{--- 1}$$

$$\mu^2 = \frac{2 + \dim(g)}{2 \cdot 5!} \left(\frac{i}{2\pi} \right)^2 \quad \text{--- 2}$$

$$\lambda_{g\mu} = \frac{-2h^\vee}{4!} \left(\frac{i}{2\pi} \right)^2 \quad \text{--- 3}$$

We have derived ① and ② already.

③ comes from cancelling mixed gauge-gravitational anomalies.

Remarkably ③ is consistent with ① and ②.

This could be possible only because the η field runs in the loop and changes the purely gravitational anomaly from

$$\frac{\dim(g)+1}{20} \int \text{Ch}_4(T_{\mathbb{P}^3}) \quad \text{to} \quad \frac{\dim(g)+2}{20} \int \text{Ch}_4(T_{\mathbb{P}^3})$$